PAPER_197: F_U_Bi_i Extended Integral — UV, mm-Wave, Hybrid, and Hierarchical Terms

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$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57$$

$$U_{b_i}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57,\ \beta_i = 0.61$$

Abstract

This paper documents the extended form of the F_U_Bi_i buoyancy integral, incorporating four new terms beyond the standard UQFF formulation: F_UV (GALEX/Spitzer UV flare coupling), F_mm (ALMA mm-radio coupling), F_hyb (hybrid polarization-frequency term), and F_hier (hierarchical remnant unification). Numerical coefficients k_UV = k_mm = 10?³° N/W and f_mm = 1.05 are derived from observational calibration. This extended integral enables multi-wavelength coupling of buoyancy forces from UV through millimeter radiation fields.

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1. Standard F_U_Bi_i Integral (Baseline)

The standard buoyancy integral form:

```
F_U_Bi = -F0 + (m_e c²/r²)·DPM_momentum·cos? + (GM/r²)·DPM_gravity + F_U_Bi_i
```

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2. Extended F_U_Bi_i Integral

The complete extended form including all observational coupling terms:

```
F_U_Bi_i = ?0^{x2} [
    -F0
  + (m_e c²/r²) · DPM_mom · cos?
  + (GM/r²) · DPM_grav
  + ?_vac,[UA] · DPM_stab
  + k_LENR · (?_LENR/?0)²
  + k_act · cos(?_act · t)
  + k_DE · L_X
```

```
  + 2qB0V · sin? · DPM_res · P_pol
  + k_neutron · s_n
  + k_rel · (E_cm,astro,enhanced/E_cm)²
  + k_UV · L_UV           ? NEW: UV flare coupling
  + k_mm · L_mm · f_mm   ? NEW: mm-radio coupling
] dx
```

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3. New Extended Terms

3.1 F_UV — GALEX/Spitzer UV Coupling

```
F_UV = k_UV · L_UV

Parameters:
  k_UV = 10?³° N/W    (UV force coupling constant)
  L_UV = UV luminosity (W) from GALEX or Spitzer photometry
  Physical origin: UV flare irradiation pressure on buoyant plasma shells
```

3.2 F_mm — ALMA mm-Radio Coupling

```
F_mm = k_mm · L_mm · f_mm

Parameters:
  k_mm = 10?³° N/W    (mm-radio force coupling constant)
  L_mm = mm-radio luminosity (W) from ALMA observations
  f_mm = 1.05          (mm-radio frequency enhancement factor)
 Physical origin: millimeter-wave radiation pressure in molecular cloud outflows
```

3.3 F_hyb — Hybrid Polarization-Frequency Term

```
F_hyb = P_pol · f_mm · (?0)?¹

Parameters:
  P_pol = polarization fraction (dimensionless, typically 0.01–0.1)
  f_mm = 1.05
  ?0 = base UQFF angular frequency (rad/s)
 Physical origin: coupling of polarized mm-emission to buoyancy oscillation frequency
```

3.4 F_hier — Hierarchical Remnant Unification

```
F_hier = S? (v_i/c)^n · ?0^{-m}

Parameters:
  v_i = velocity of remnant component i (m/s)
  c = speed of light
  n = 2 (hierarchical power index)
  m = 1 (frequency suppression exponent)
```

Physical origin: multi-component remnant velocity hierarchy (e.g., jet + cocoon + lobe)

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4. Standard Integral Terms (Reference)

Term	Symbol	Physical Origin
Base restoring force	-F0	Vacuum restoring force
DPM momentum	$(m_e c^2/r^2)$·DPM_mom·cos?	Dipole-plasma momentum scattering
DPM gravity	(GM/r^2)·DPM_grav	Dipole-plasma gravitational coupling
DPM stability	?_vac,[UA]·DPM_stab	Vacuum aether stability term
LENR coupling	k_LENR·$(?\_LENR/?0)^2$	Low-energy nuclear resonance
Activation term	k_act·cos(?_act·t)	Activation oscillation
Dark energy luminosity	k_DE·L_X	X-ray dark energy coupling
DPM resonance	2qB0V·sin?·DPM_res·P_pol	Magnetic resonance polarization
Neutron cross-section	k_neutron·s_n	Neutron scattering
Relativistic CM	k_rel·$(E\_cm,astro,enh/E\_cm)^2$	Relativistic center-of-mass enhancement

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5. Numerical Results for Extended Terms

System	Standard F_U_Bi_i	With UV/mm extension
Magnetar SGR1745	˜ $2.11\times10^{2°8}$ N	+ F_UV from Chandra/GALEX
NGC 3603	˜ -8.31×10^{211} N	+ F_mm from ALMA CO observations
Pillars of Creation	˜ $9.79\times10?^{33}$ N	+ F_hyb with P_pol ˜ 0.05

Note: The extreme magnitudes ($10^{2°8}$ N, 10^{211} N) reflect vacuum-density-scaled units in the UQFF framework where ?_vac,[UA] ˜ $10?^{113}$ (dimensionless normalized).

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6. Integration with Existing UQFF Terms

The extended integral fits within the Triadic Master System (PAPER_196) as the primary buoyancy channel FU_Bi. Specifically:

- **F_UV** activates when GALEX UV flux > threshold (flare events)
- **F_mm** activates when ALMA observes SFR-driven mm continuum
- **F_hyb** operates continuously as polarization coupling
- **F_hier** applies to multi-component systems (AGN jets, SNR shells)

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7. References

- `grok_share_7514fe.txt` lines 200–400 (first PDF extended integral section)
- PAPER_196: Triadic Master Equation System
- PAPER_171: Universal Gravity Ug1–Ug4 Decomposition
- PAPER_172: FU Complete Unified Field Assembly